

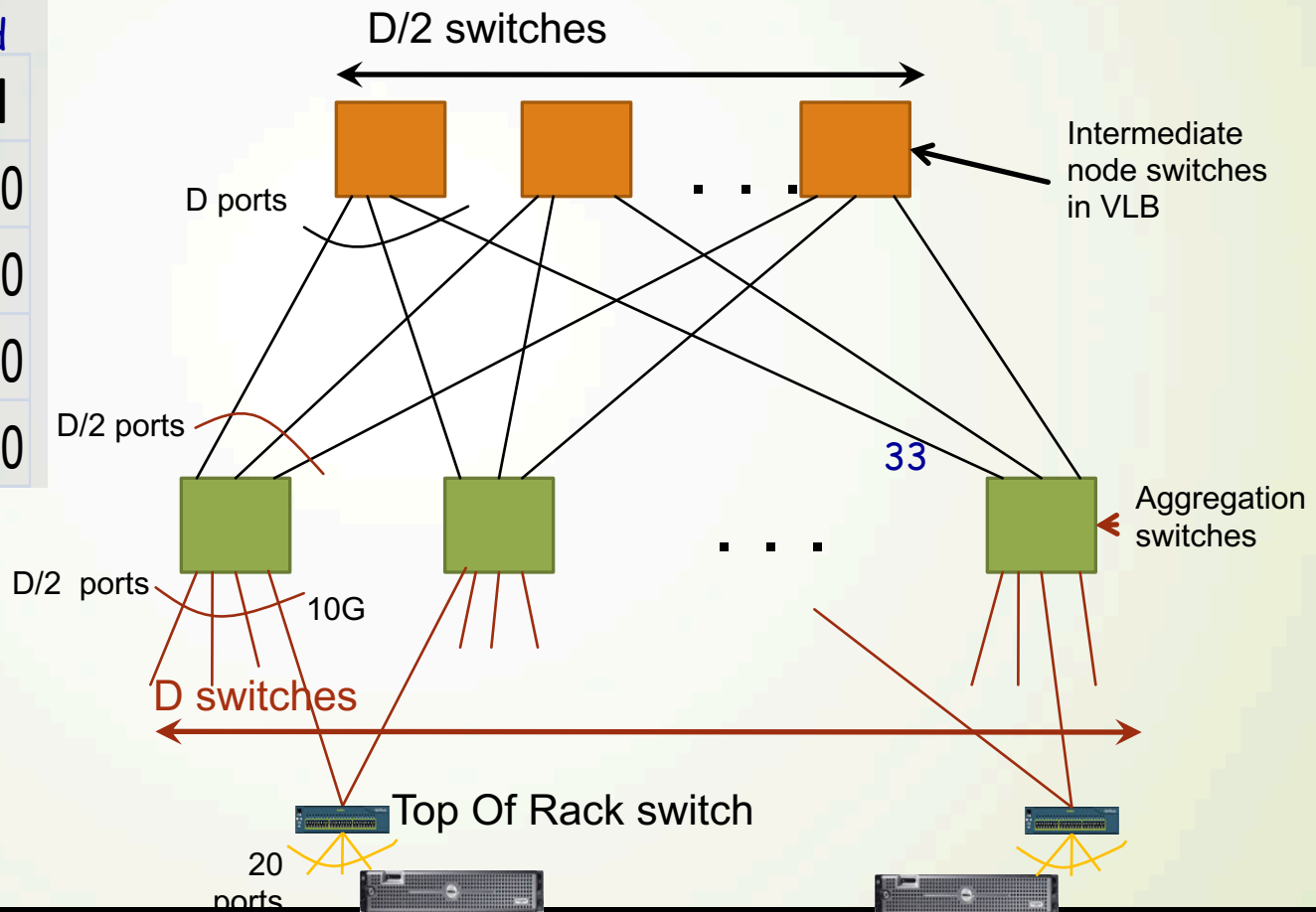
VL2 Topology Design: Scale-out vs. Scale-up

- Exploit the gap in **switch-to-switch** capacity vs. **switch-to-server** capacities
 - current: 10Gbps vs. 1Gbps; future: 40 Gpbs vs. 10 Gbps
- A scale-out design with broad layers
 - E.g., a 3-level Clos topology with full-mesh in top-2 levels
 - ToR switches, aggregation switches & intermediate switches
 - **less wiring complexity, and more path diversity**
- Same bisection capacity at each layer → no **oversubscription**
- Extensive **path diversity** → graceful degradation under failure

Sample VL2 Topology

Node degree (D) of
available switches &
servers supported

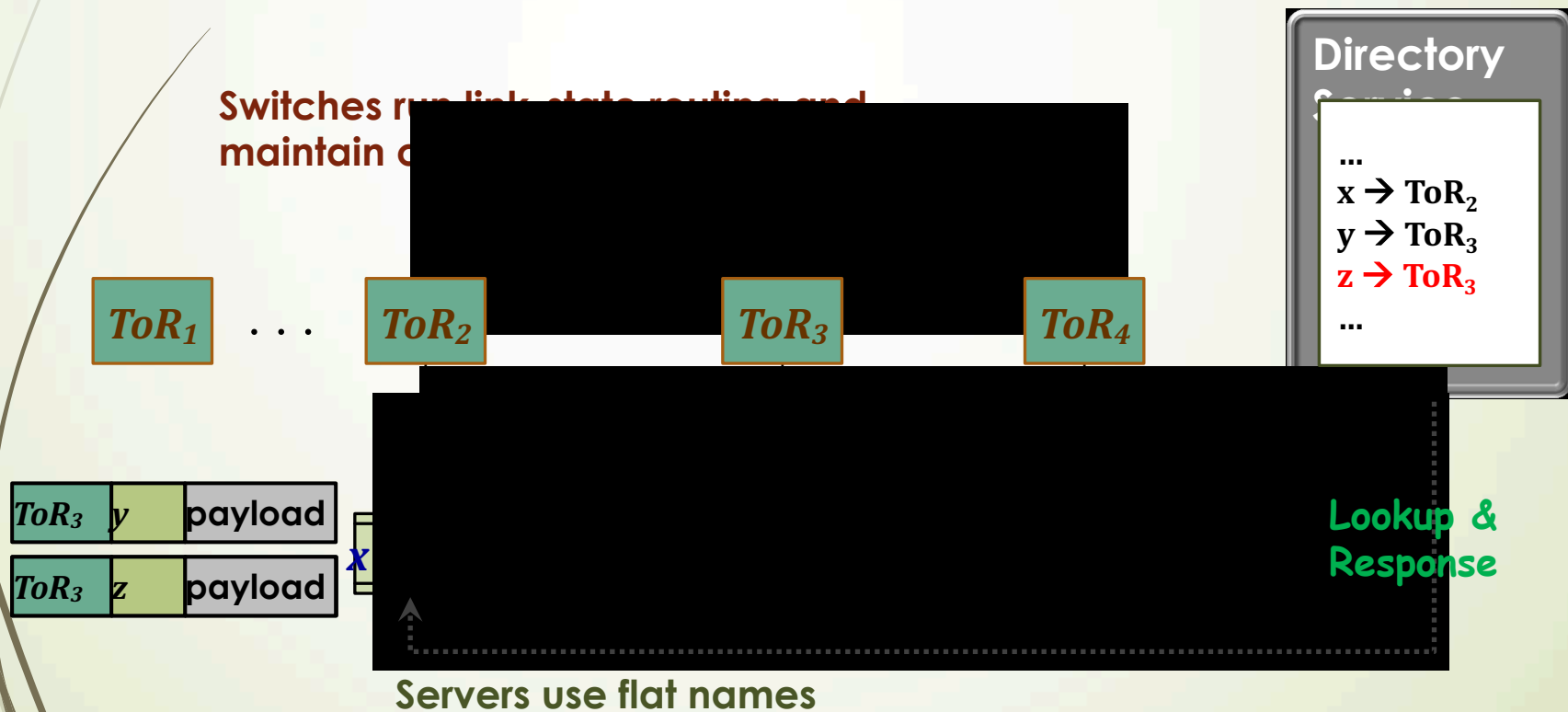
D	# Servers in pool
4	80
24	2,880
48	11,520
144	103,680



Addressing and Routing in VL2

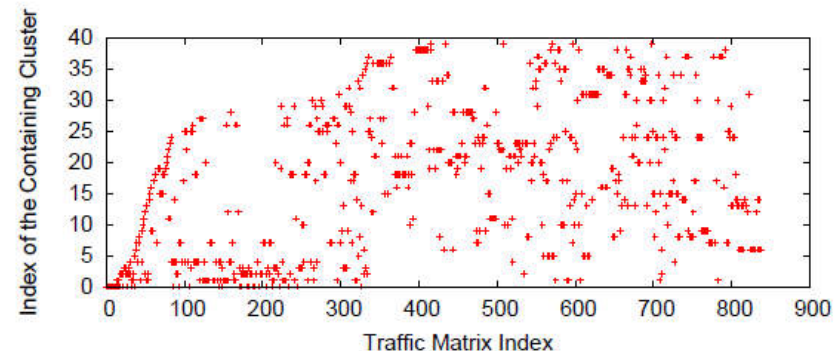
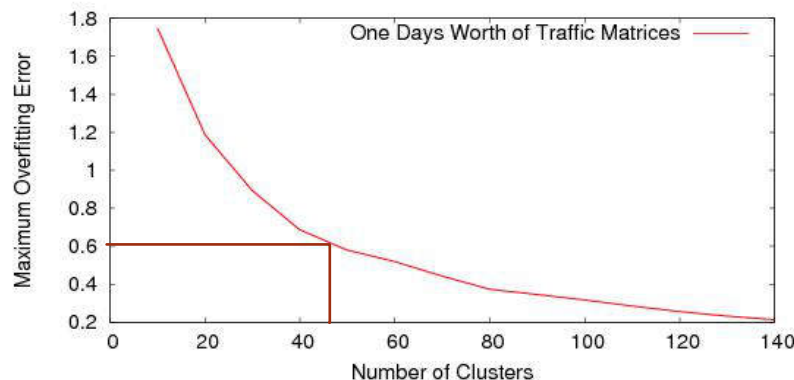
- Allows to use low-cost switches
- Protects network and hosts from host-state churn
- Obviates host and switch reconfiguration

Switches run link state routing and maintain a



Traffic Matrix Measurements: No Predictivity!

- Collapse similar traffic matrices (over 100sec) into “clusters”
- Need 50-60 clusters to cover a day's traffic
- Run length is 100s to 80% percentile; 99th is 800s
- Traffic pattern changes nearly constantly



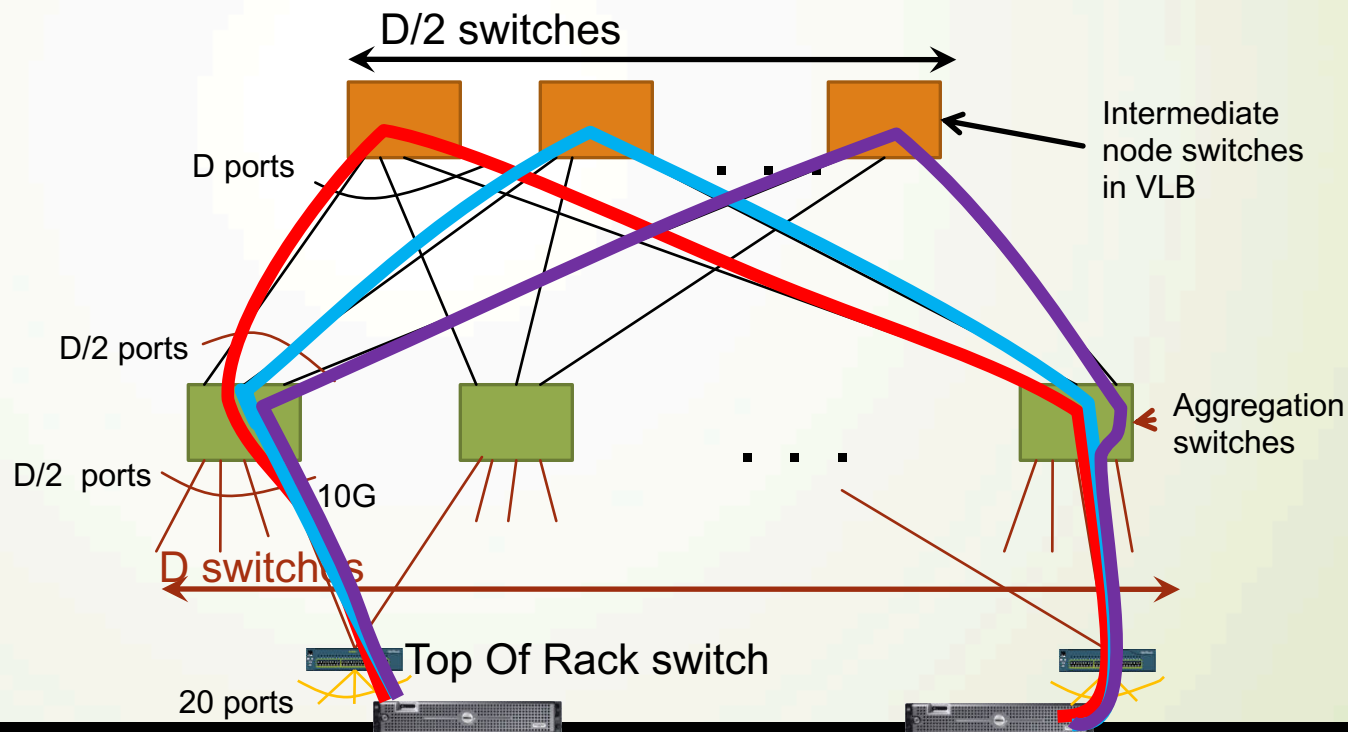
Randomization to Rescue

Valiant Load Balancing

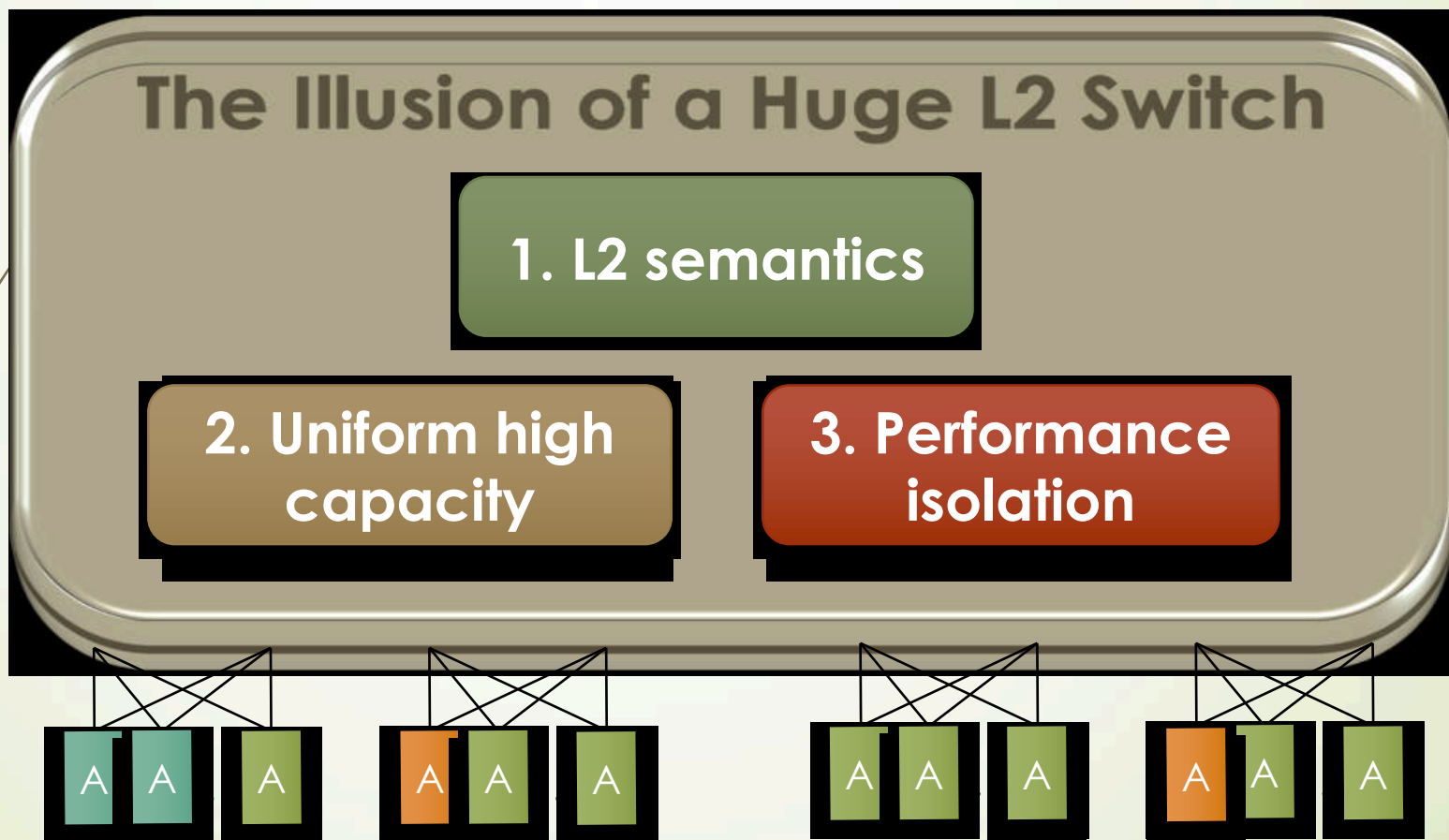
- Every flow **bounced** off a random intermediate switch
- Provably** hotspot free for any admissible traffic matrix

Node degree (D) of available switches & # servers supported

D	# Servers in pool
4	80
24	2,880
48	11,520
144	103,680



VL2 Achieves Agility at Scale Via....



Take-Away Message

- Main Question for Data Center Networks:
 - What are the unique/new networking issues posed by large-scale data centers?
- We have seen
 - Network Architecture?
 - Topology design?
 - Addressing?
 - Routing?
 - Forwarding?
 - Anything else?

Questions

chsu@cs.nthu.edu.tw

